Diego Cuevas Vega

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EDUCATION

Harvard College Cambridge, MA

S.B. in Electrical Engineering, Secondary Field in Economics Expected May 2029

Relevant Coursework: Introduction to Digital Fabrication; Circuits, Devices, and Transduction; Systems and Control

Honors:

- Jacob Wendell Scholarship Prize Invitee (among the top first-year academic records at Harvard College), 2026
- Gold Medal, International Economics Olympiad (IEO), 2025 — first-ever Mexican gold medalist
- Gold and Bronze Medals, Mexican Mathematical Olympiad Championships (two editions), 2024

TECHNICAL SKILLS & CERTIFICATIONS

Robotics & Embedded: ROS2, C/C++ (Arduino), Python, Docker, Linux, ESP32, XIAO, I2C, UART, IMU, RTK GPS

Hardware & Electronics: Soldering, oscilloscope, multimeter, SOLIDWORKS, Fusion 360, Onshape, 3D printing

Certifications: SOLIDWORKS Associate (CSWA); Harvard SEAS/Physics Machine Shop Certification (mill and lathe)

Languages: Spanish, English, French (Proficient, DELF B2), German (Upper-Intermediate, Goethe-Zertifikat B2)

EXPERIENCE

Harvard University, Department of Physics Cambridge, MA

Course Assistant, FYSEMR 58E: Machine Ecology (Autonomous Robots for Restoration) Sep 2026 – Present

- Integrated RTK GPS positioning into the rover's navigation stack for autonomous waypoint following
- Co-developed and debugged a pure pursuit path-tracking controller fusing RTK GPS and IMU data
- Co-designed electronics enclosures and end-effector mechanisms in CAD; converted geodata into rover waypoints
- Lead 12 students across navigation, mechanical, and soil/ecology subteams under the faculty instructor

Harvard University, Digital Fabrication Laboratory Cambridge, MA

PRISE Research Fellow, Robotics for Ecological Restoration Jun 2026 – Aug 2026

- Designed two rover-agnostic ~$330 berm-digging end effectors, field-tested on Jackal and RMP 440 rovers
- Wrote ESP32-C3 actuator firmware with a motor driver and built ROS2 nodes communicating over WebSockets
- Ran berm compression tests, identifying a 1.8 A slippage threshold and a method with up to 3x wall pressure

PROJECTS

Oscillating Air Engine, *Harvard Machine Shop Certification* Jul 2026

- Machined all 11 parts of an oscillating engine from steel, brass, aluminum, and acrylic on a manual mill and lathe
- Worked from engineering drawings with tolerances down to ± 0.0005 in; drilled and tapped 4 thread sizes

UNDP Philippines Budget Dashboard, *Harvard Global Research & Consulting (GRC)* Jan 2026 – May 2026

- Built a comparison tool that lets UNDP economists analyze Philippine national budget tables side by side
- Developed climate change expenditure views by department and NCCAP strategic priority, with supporting analysis
- Iterated on features with a UNDP country economist; presented to their office, where it is now in active use

Wireless Bike Turn-Signal System, *PS70: Introduction to Digital Fabrication* Sep 2025 – Dec 2025

- Built a wireless bike turn-signal system linking a handlebar XIAO to a rear ESP32 driving NeoPixels via ESP-NOW
- Soldered LiPo power systems with buck/boost converters into space-constrained 3D-printed enclosures

EXTRACURRICULARS

Project Lead, *Harvard Undergraduate Clean Energy Group (HUCEG) BUILD* Jan 2026 – Present

- Lead 10 members building a solar-powered hive monitor for weight, bee count, temperature, humidity, and sound
- Selected the sensor suite and microcontroller; now integrating and bench-testing each sensor for the first prototype

Program Director, *PRIMEX (Program to Increase Mexican Representation at Harvard)* Sep 2026 – Present

- Doubled mentors to 28 across 9 Harvard schools for a free college admissions program for Mexican applicants
- Drew 150+ applications for 50 spots by leading outreach and media, including a video with 400K+ views

Sourcing Analyst, *Harvard Undergraduate Capital Partners (HUCP)* Sep 2025 – Present

- Source ~5 startups a week for a partner VC firm; recommendations have produced 3 warm intros to founders

Board Member, *Harvard University Mexican Association (HUMEX)* Sep 2026 – Present

- Plan and budget weekly programming for 20+ members, strengthening Harvard's Mexican student community

PUBLICATIONS

Toluwaniase Ademola, R. McCarthy, J. Gokongwei, **Diego Cuevas Vega** J. R.M. Kennedy, K. L. Bar-Sinai, N. Melenbrink, "Assessing the Potential Utility of Low-Cost Robotic End-Effectors for Ecological Restoration," *IEEE GHTC*, 2026.